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VideoEspresso: A Large-Scale Chain-of-Thought Dataset for Fine-Grained Video Reasoning via Core Frame Selection

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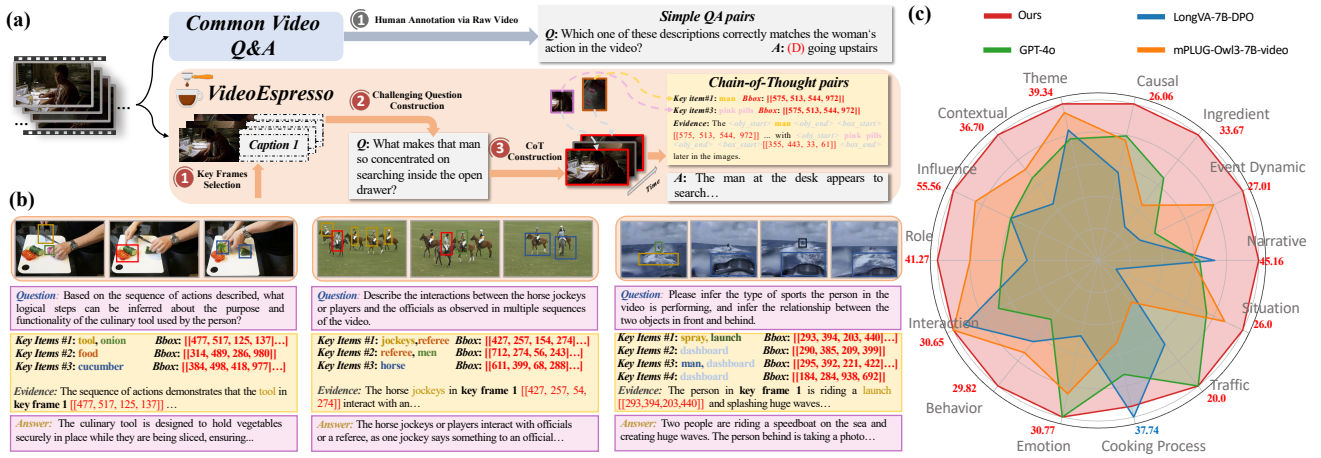

Figure 1. **Overview of *VideoEspresso*.** **(a)** Comparison of annotation pipelines: Unlike traditional videoQA datasets, *VideoEspresso* features an automatic pipeline for constructing complex reasoning QA tasks and multimodal Chain-of-Thought (CoT) annotations. This enhances the diversity of QA data and significantly improves scalability. **(b)** Examples from *VideoEspresso*: Illustrated are sample question-answer pairs, along with CoT bounding boxes and evidence annotations, demonstrating the dataset's richness. **(c)** Benchmark performance: Comparative results on our benchmark highlight the video reasoning capabilities of our model.

Abstract

The advancement of Large Vision Language Models (LVLMs) has significantly improved multimodal understanding, yet challenges remain in video reasoning tasks due to the scarcity of high-quality, large-scale datasets. Existing video question-answering (VideoQA) datasets often rely on costly manual annotations with insufficient granularity or automatic construction methods with redundant frame-by-frame analysis, limiting their scalability and effectiveness for complex reasoning. To address these challenges, we introduce VideoEspresso, a novel dataset that features VideoQA pairs preserving essential spatial details and temporal coherence, along with multimodal annotations of intermediate reasoning steps. Our construction pipeline employs a semantic-aware method to reduce redundancy, followed by generating QA pairs using GPT-4o. We further develop video Chain-of-Thought (CoT) annotations to enrich reasoning processes, guiding GPT-4o in extracting logical relationships from QA pairs and video content. To exploit the potential of high-quality VideoQA pairs, we propose a Hybrid LVLMs Collaboration framework, featuring a Frame Selector and a two-stage instruction fine-tuned reasoning LVLM. This framework adaptively selects core frames and performs CoT reasoning using multimodal evidence. Evaluated on our proposed benchmark with 14 tasks against 9 popular LVLMs, our method outperforms existing baselines on most tasks, demonstrating superior video reasoning capabilities. Our code and dataset have been released at: https://github.com/hshjerry/VideoEspresso

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1. Introduction

In recent years, the development of Large Vision Language Models (LVLMs) [3, 11, 20, 26] has brought significant advancements to multi-modal understanding tasks. By integrating visual and language information through extensive data training, more advanced LVLMs families [17, 33] are able to generate reasonable outputs while fully leveraging the rich knowledge of LLMs, thus demonstrating excellent performance in tasks like image captioning and visual question answering. Recent research has started to explore extending LVLMs to the domain of video content understanding [21, 24, 29, 37]. Although these efforts have shown great potential on certain basic video understanding benchmarks [22, 30, 46, 52], their performance on complex video reasoning tasks remains less than satisfactory.

A primary limitation in video question-answering (VideoQA) research is the scarcity of high-quality, large-scale datasets. Current VideoQA datasets [16, 45, 52, 57] rely on costly manual annotations that often lack the granularity needed for detailed understanding, limiting scalability. Yet, LVLMs require vast amounts of multimodal QA pairs for effective training. Recently, advancements in large language models (LLMs) like GPT-4 [32] and Gemini-Pro [12] have allowed for the automatic generation of QA pairs through carefully designed prompts. A straightforward approach is to use video metadata—typically high-level descriptions—and leverage LLMs to generate QA pairs based on this coarse information. However, the missing of crucial video details restricts the QA pairs' effectiveness for fine-grained reasoning. Alternatively, analyzing video frames for a more granular understanding is feasible, but video content is often redundant, with key information dispersed sparsely, making frame-by-frame analysis computationally expensive and prone to information overload.

To address these challenges, we propose a novel automatic VideoQA construction method and introduce a new dataset, *VideoEspresso*. By preserving important spatial details with temporal coherence, we create a fine-grained reasoning-enabled VideoQA dataset that fosters more effective multimodal understanding. As shown in Fig. 1, we first design a semantic-aware key information extraction method to extract key information from the video. Unlike traditional methods that extract key frames based on image representations, we first map video frames to the linguistic space using an LVLM. We then remove similar frames based on semantic similarity, which reduces redundancy in the video data. To retain frame-level details and inter-frame correlation, we sequentially group the video frames and input them into GPT-4o [33]. With carefully designed prompts, we instruct the model to generate initial QA pairs and filter out low-quality data. To further expand the intermediate reasoning steps, we introduce video Chain-of-Thought annotations. We design prompts to guide GPT-4o in extracting logical relationship evidence from QA pairs and videos that are helpful for answers, including interactions of key objects in spatial and temporal flow. By annotating these logical processes, we ultimately aim to expand the chain of inference evidence on QA datasets.

To fully leverage the potential of the high-quality VideoQA pairs in our proposed *VideoEspresso*, we introduce a novel framework, Hybrid LVLMs Collaboration for VideoQA, achieving cost-effective and accurate video LVLM reasoning. The framework is composed of a tiny *Frame Selector* and a fine-grained reasoning LVLM. The *Frame Selector* adaptively selects the most relevant core frames for the question based on the image-to-language mapping. These core frames are then submitted to the reasoning LVLM, where the model first extracts multi-modal evidence based on the frame information, and ultimately provides the answer to the question through chain-of-thought reasoning, leveraging this evidence. This dataset provides explicit annotations of key reasoning steps and image regions through both text and bounding boxes, which enables models to effectively use text and image localization information when answering questions.

Based on our dataset, we have constructed an evaluation benchmark that includes a set of GPT-4o-based open-ended evaluation metrics. We evaluated 9 popular LVLMs as our comparison baselines. To assess video reasoning capabilities from different perspectives, we categorized the evaluation into 14 tasks. Our method demonstrates significant advantages over baseline methods across most tasks.

2. Related Work

VideoQA Dataset. Traditional videoQA datasets [16, 46, 52] rely heavily on manual annotations, where annotators watch videos, summarize content, and generate QA pairs based on set guidelines. This labor-intensive process limits scalability. Moreover, recent study [48] has effectively extended videoQA to the embodied AI scenario, highlighting the importance of fine-grained spatial understanding. With advancements in LLM capabilities [1, 12, 31], recent approaches use tailored prompts to utilize LLMs for annotation, often relying on metadata or detailed captions [6, 39] to construct QA data [22, 35, 38]. However, these methods often lack fine-grained video information and depend on raw video data. In contrast, we introduce an automatic pipeline for QA pair generation that processes and annotates raw data without manual input, enhancing scalability.

Video LVLMs. In recent years, large vision-language models (LVLMs) have advanced VideoQA tasks significantly. Prior works have enhanced model performance by experimenting with various architectures, with some aligning visual and textual features through Q-Former [5, 17, 41, 53], while others concatenate frame-level features directly [14, 15, 21]. Balancing frame count with token efficiency has

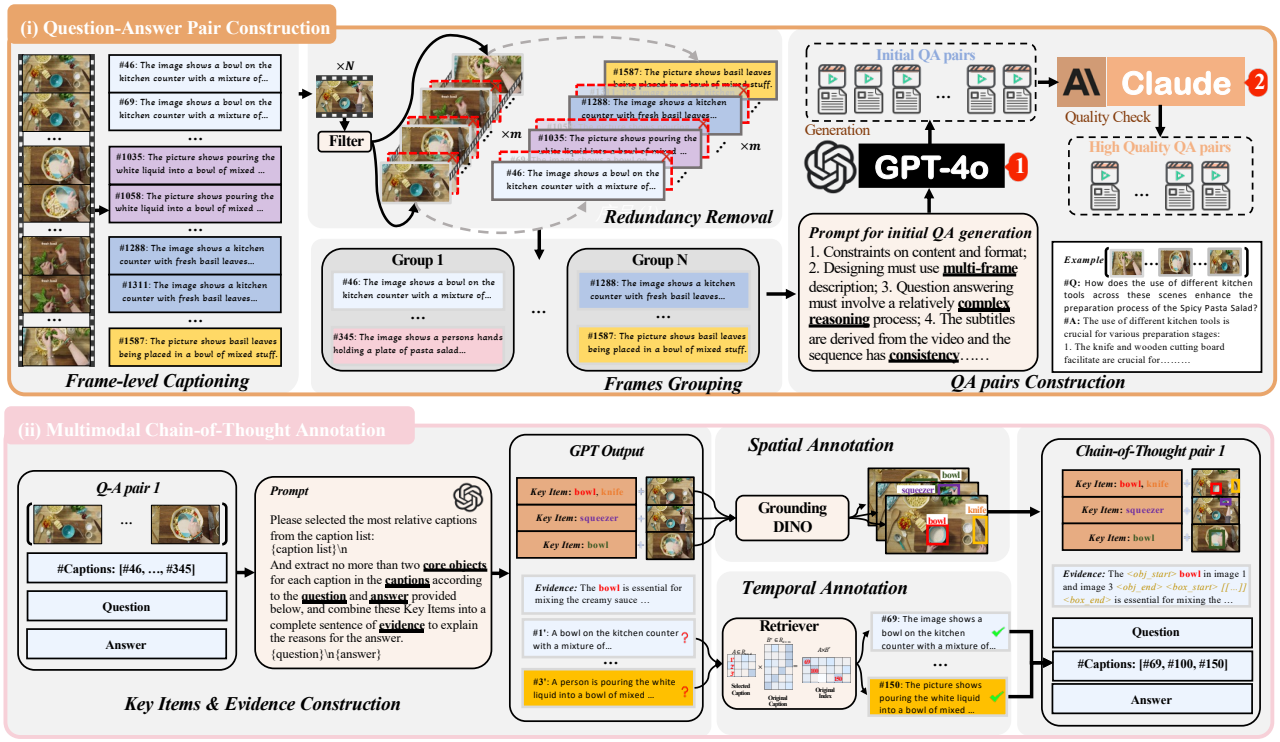


Figure 2. **The automatic generation pipeline of *VideoEspresso*.** (i) **Question-Answer Pair Construction**: We use video frame-leveled captions to extract the key frames of the video and group descriptions of these frames. Then, we prompt GPT-4 to design questions for each group of video frames. (ii) **Multimodal Chain-of-Thought Annotation**: We extract key evidence text and generate captions with the highest relevance to the question with GPT-4o. Additionally, we annotate spatial and temporal information for key items, which results in multimodal Chain of Thought data pairs grounded in both temporal and spatial dimensions.

become a key focus, typically addressed by uniform sampling [5, 54] or additional modules [19, 24, 38]. Our approach employs a tiny model that selectively captures frames relevant to the question, minimizing context length while preserving essential spatiotemporal information.

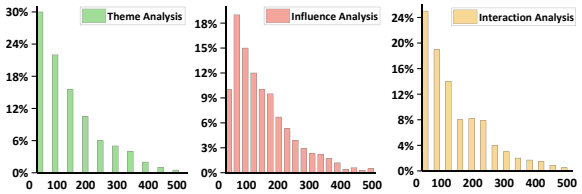
Visual CoT. Chain-of-Thought (CoT) techniques enhance the reasoning abilities of LLMs by guiding them through intermediate reasoning steps to produce more accurate answers. While prior works [50, 56] have applied CoT to visual tasks, they primarily focus on text-level reasoning, often overlooking visual comprehension. Recent studies [36, 43] improved performance by targeting specific image regions. In videoQA, VideoCoT [42] focuses on text-level reasoning, while VoT [10] emphasizes spatial relationships. Our approach integrates key object regions and core frames in the video CoT reasoning, capturing both spatial and temporal details to enhance video understanding.

3. VideoEspresso

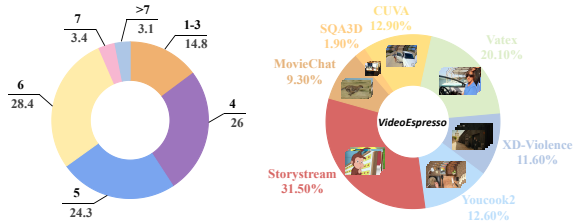
In this section, we introduce *VideoEspresso*, a large-scale VideoQA dataset designed to facilitate high-level reasoning over macroscopic video semantics. This dataset is generated through a scalable, fully automated generation pipeline that produces high-quality reasoning VideoQA pairs from distilled video content. The *VideoEspresso* construction pipeline consists of the following key stages: (1) We collect raw video data, followed by redundancy reduction streamline video frames that encapsulate essential content; (2) Based on these frames, we generate QA pairs that capture the core semantics of each video; (3) To further enable interpretability and bolster the benchmark for complex reasoning capabilities, we incorporate fine-grained Chain-of-Thought (CoT) annotations, which connect core visual elements through spatial and temporal interactions, bridging the reasoning gaps in traditional VideoQA pairs.

3.1. Video Data Curation

We leverage the vast amount of unannotated Internet videos for building scalable datasets. To build a videoQA dataset with complex semantic reasoning, selecting appropriate data sources and types is essential. As illustrated in Fig. 3, we collect raw videos from 7 datasets [9, 28, 38, 40, 44, 49, 58] encompassing rich temporal dynamics, logical sequences, and causal relationships. These high-level semantics provide a strong foundation for constructing a complex and coherent question-answering dataset. We conduct a manual review of these videos to evaluate their reasoning



(a) **Distribution of Distance** between adjacent core frames



(b) **Number of key items** (c) **VideoEspresso Data Source**

Figure 3. **The statistical analysis of our *VideoEspresso* dataset.**

potential. The dataset includes diverse video types, covering genres such as news, movies, documentaries, animation, and educational content. Based on these characteristics, we predefined 14 tasks to assess the model's capabilities across various dimensions of video reasoning tasks.

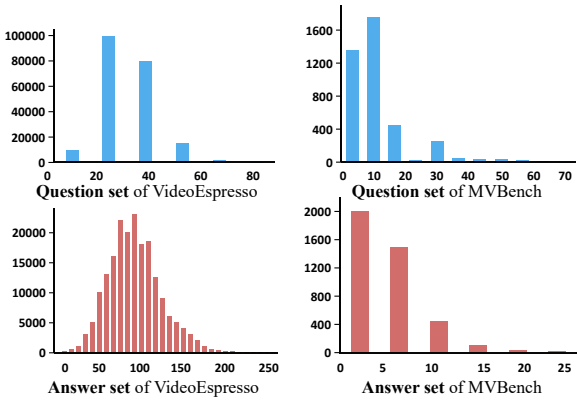
3.2. Redundancy Removal in Video Frames

The goal of this module is to eliminate redundant information in the video and retain the essential content by selecting a concise sequence of frames. Different videos exhibit varying rates of content and scene changes. Initially, we determine an appropriate sampling interval based on the type of video. For instance, for rapidly changing dynamic scenes, we set the FPS between 2 and 4, whereas for static scenes, we choose a longer sampling interval, with FPS set to 1. Then, to capture the fine-grained semantic information of the video as input for constructing QA pairs, we use InternVL2-8B [7] to perform frame-level captioning on all sampled frames. To filter out redundant frames in the video, we leverage the language retrieval model BGE-M3 [4] to preliminarily remove highly similar frames through fine-grained semantic filtering. Specifically, for all sampled frame descriptions $\mathcal{C}$, if the cosine similarity between the textual features $\phi_T(c)$ of adjacent captions exceeds a preset threshold τ, we apply a Last-In-First-Out (LIFO) filtering approach. This process results in a concise caption sequence $\mathcal{C}'$ and the corresponding frames. The process is formulated as follows:

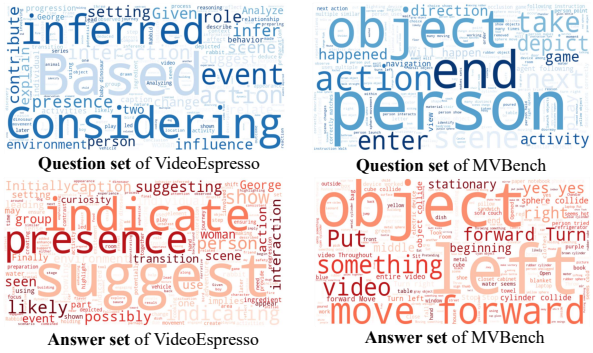
$$\mathcal{S} = \arg\max_{c_i, c_j \in \mathcal{C}} \cos(\phi_T(c_i), \phi_T(c_j)), \tag{1}$$

$$\mathcal{C} \longrightarrow \mathcal{C}' \ (c \in \mathcal{C}', \text{if } \mathcal{S}(c) < \tau), \tag{2}$$

where $\mathcal{S}$ denotes the similarity matrix, and $\phi_T(c)$ represents the feature computation for the caption c.



(a) Comparison of **Token Length**



(b) Comparison of **Word Cloud**

Figure 4. **The dataset attributes comparison between our *VideoEspresso* and MVbench.**

3.3. Question-Answer Pair Construction

This module aims to leverage the powerful language reasoning capabilities of LLMs to automatically construct high-quality video reasoning QA pairs based on detailed descriptions of video frames. To maintain semantic continuity within the groups and avoid issues such as model hallucinations and failure to follow instructions—caused by an excessive number of tokens—we adopt a continuous grouping approach to streamline frames. Specifically, for all captions $\mathcal{C}'$ of a single video, every 15 consecutive frame captions are grouped into a group $\mathcal{G}_i$ that preserves both frame-level details and inter-frame correlations.

To obtain complex reasoning QA pairs, we design and iteratively refine prompts to ensure that the LLM follows the rules in constructing them. As shown in the top-right corner of Fig. 2, our prompts instruct GPT-4o [33] to generate question-answer pairs based on multi-frame descriptions, while also ensuring that GPT-4o maintains consistency between the descriptions to construct complex video reasoning questions. To improve the quality of the QA pairs, we validate them by designing an additional LLM [1] to verify the quality of both the questions and answers, including eliminating hallucinations in the QA pairs, checking the factual accuracy of the answers, and filtering out answers to

highly subjective or difficult-to-evaluate open-ended questions. Ultimately, we obtain high-quality reasoning QA pairs for these videos and record the frame sequence grouping corresponding to each QA pair.

3.4. Multimodal Chain-of-Thought Annotation

To further enhance the reasoning capabilities of models, this module focuses on annotating multimodal evidence that contains key spatiotemporal information. First, we group the Q-A pairs obtained in Sec. 3.3 along with the corresponding frame sequences as input, and design the prompt shown in the lower-left corner of Fig. 2 to guide GPT-4o [33] in extracting key information. Sparse core frames are sufficient to capture enough information to answer the question; hence, we need to obtain the core frames most relevant to the question. Accordingly, our designed prompt primarily guides GPT-4o to extract the following key information: (1) From the captions group, select the captions most relevant to the question, i.e., the captions of core frames; (2) extract key objects from these captions, i.e., the key items; and (3) organize these key objects into a natural language description as evidence to support answering the question, i.e., the evidence. To expand the dimensions of reasoning, we annotate these key elements with temporal and spatial information. For spatial annotation, we apply GroundingDINO [27] to mark bounding boxes around all key items and leverage CLIP-ViT-B/32 [34] to verify consistency between labels and objects within the bounding boxes. For temporal annotation, since GPT-4o's generated captions of core frames $\mathcal{G}_{GPT}$ in (1) cannot directly match the original captions at the string level, we employ BGE-M3 [4] to retrieve the caption in the original set $\mathcal{G}_i$ and obtain temporal grounding information t. The process is formulated as follows:

$$t = \arg\max_{k} \cos(\phi_T(c_j), \phi_T(c_k)), \tag{3}$$

where $c_j \in \mathcal{G}_{GPT}$, $c_k \in \mathcal{G}_i$.

Finally, we obtain both textual evidence and multimodal key information that includes temporal and spatial dimensions, which can serve as the intermediate reasoning step to assist in answering questions.

3.5. Data Analysis

In Fig. 3 and Fig. 4, we provide a visualization of the data analysis and comparisons. To investigate the temporal distribution of key information in videos, we first examine the distribution of distances between adjacent key frames across different tasks. As shown in Fig.3.(a), the distance distribution between key frames varies significantly across tasks, indicating that the conventional strategy of uniformly sampling video frames is suboptimal and introduces substantial redundancy. As shown in Fig. 3.(b), the number of key items in the CoT of our dataset is diverse, encompassing a range from a few to numerous critical elements, reflecting the complexity and diversity of the visual content. In addition to the unique characteristics of our dataset, we also compare it with the QA content of a popular dataset, MVBench [22]. As shown in Fig. 4.(a), we illustrate the differences between the datasets in terms of token length. The QA set length in MVBench (right) is shorter, while the answer set in our *VideoEspresso* (left) is on average much longer and exhibits greater diversity in distribution. As illustrated in Fig. 4.(b), we further present a comparison of the word clouds for *VideoEspresso* and MVBench. In the Question set, our *VideoEspresso* emphasizes reasoning based on visual facts, with keywords like "*considering*", "*based*" and "*inferred*" In contrast, MVBench emphasizes basic inquiries such as "*object*", "*person*" and "*action*" In the Answer set, *VideoEspresso* includes not only reasoning-related keywords as previously described but also terms associated with reasoning steps, such as "*Initially*" and "*Finally*" On the other hand, MVBench focuses on object definitions and spatial relationships within videos, with keywords like "*object*", "*left*" and "*forward*".

4. Hybrid LVLMs Collaboration for VideoQA

To fully unleash the potential of high-quality video QA pairs offered by *VideoEspresso*, we propose an efficient video reasoning framework with hybrid LVLMs collaboration to enable cost-effective and accurate video LVLM reasoning. As shown in Fig. 5 the framework consists of two core components: a lightweight selector that identifies core frames that are closely related to the input question, and a powerful LVLM that performs content understanding and reasoning based on these selected core frames.

4.1. Core Frames Selection via Tiny LVLM

We propose a Lightweight Selector designed to extract core frames that are closely aligned with the question from input videos. Unlike traditional keyframe extraction methods, which primarily filter out semantically similar frames, our approach dynamically selects question-driven core frames to meet diverse task requirements. This enables a reduction in the number of frames passed to large models compared to conventional methods that rely solely on frame semantic similarity. Moreover, this selector serves as a plug-and-play module that can be inserted before any LVLMs.

Our architecture comprises a lightweight LVLM with 1 billion parameters and an LLM with 0.5 billion parameters in a sequential setup. The LVLM's function is to convert video frames into language descriptions, while the LLM selects frames most relevant to the question based on these descriptions. Specifically, the process consists of two steps.

(1) Frame Captioning: Given a video V and a specified frames-per-second (`FPS`) sampling rate, the `LVLM` samples

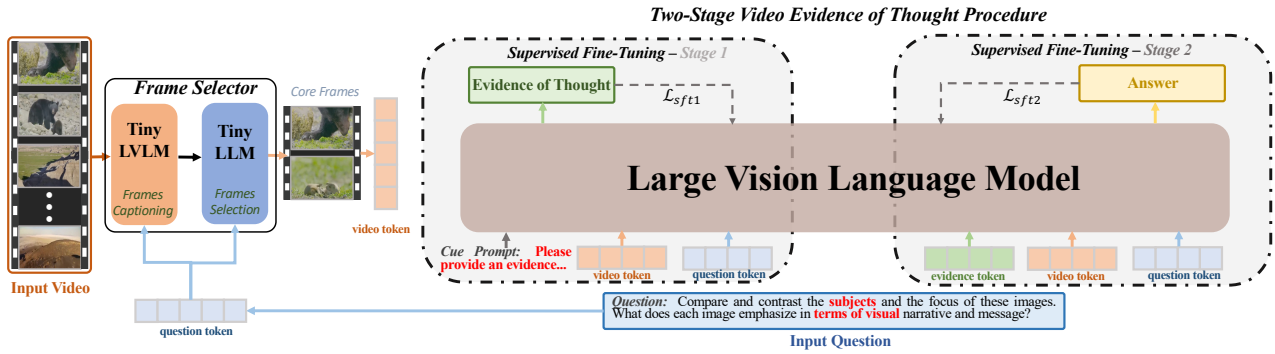


Figure 5. **Two-Stage Video Evidence of Thought Training Procedure.** The Frame Selector comprises a tiny LVLM and a tiny LLM, tasked with generating captions for videos and selecting the most relevant frame to as core video token for large reasoning model. A two-stage supervised fine-tuning technique is employed. During stage-1, a set of cue prompts is introduced to guide the model in producing evidence, while in stage-2, the evidence generated from stage-1 is concatenated and used directly to guide the answer generation.

frames and generates a caption c_i for each frame f_i. This process can be formulated as:

$$\{f_i\}_{i=1}^{N} = \texttt{SampleFrames}(V, \texttt{FPS}) \tag{4}$$

$$\{c_i\}_{i=1}^{N} = \texttt{LVLM}(\{f_i\}_{i=1}^{N}) \tag{5}$$

where N is the total number of sampled frames and $\{c_i\}_{i=1}^{N}$ represents the collection of captions for these frames.

(2) Core Frame Selection: Using the set of captions $\{c_i\}_{i=1}^{N}$ and a question q, the LLM identifies the subset of captions most relevant to q, resulting in a set of core frame captions $\{c'_j\}_{j=1}^{M}$:

$$\{c'_j\}_{j=1}^{M} = \texttt{LLM}(\{c_i\}_{i=1}^{N}, q) \tag{6}$$

where $M \leq N$ and $\{c'_j\}_{j=1}^{M}$represents the final set of core captions selected for their relevance to the question. Since this step requires minimal reasoning from the model, we adopt a cost-efficient solution to address the challenges posed by excessive token length when handling video input.

4.2. Fine-Grained Reasoning via LVLM

Given key frames extracted from the first stage, our goal is to enable the model to effectively leverage multimodal spatiotemporal evidence for answering complex reasoning tasks. We design a two-stage supervised fine-tuning paradigm. In the first stage, we guide the model to extract essential visual evidence from video data relevant to the question, establishing a foundation for deeper reasoning. This is achieved through supervised fine-tuning using instructions like *"Please provide evidence to help answer the question."* to guide the model in generating evidence. This evidence-based generation process not only filters core information but also enhances multimodal alignment and prepares the model for subsequent reasoning tasks.

In the second stage, we further fine-tune the model to directly generate answers based on the extracted multimodal evidence. This is achieved through supervised fine-tuning using instructions like *"Please answer the question with the help of evidence."* to guide the model in answering with evidence. Unlike traditional single-stage question-answering methods, this two-stage structure divides evidence generation and answer generation, enhancing transparency in reasoning and boosting response accuracy. Besides, it ensures that the model gradually integrates multimodal information for complex spatiotemporal reasoning, significantly improving performance in video question-answering tasks by producing more logically coherent answers.

4.3. Inference

During the inference phase, we first use the Lightweight Selector to extract core frames from the video that are closely related to the question, serving as input for subsequent reasoning. We then leverage a fine-grained reasoning LVLM, which generates evidence through a chain-of-thought process to support the final answer generation. This workflow enables efficient question answering, from frame selection to answer generation.

5. Experiments

5.1. Overview of the Evaluation Benchmark

Our *VideoEspresso* includes 14 predefined tasks, with each constructed QA pair matched to a corresponding task using GPT-4o. If no suitable task alignment is identified, the pair is categorized as "*Others*". To establish a comprehensive benchmark, the defined tasks encompass diverse perspectives, including time, logic, scene, behavior, and state, illustrated by examples such as "*Event Dynamics*", "*Causal Analysis*" and "*Theme Analysis*". Additionally, the framework incorporates real-world application tasks, such as "*Cooking Process*" and "*Traffic Analysis*". The benchmark assesses the performance of LVLMs through both ob-

Models	#Frames	Param	TFLOPs	Narra.	Event	Ingre.	Causal	Theme	Conte.	Influ.	Role	Inter.	Behav.	Emoti.	Cook.	Traff.	Situa.	Avg.
Closed-source LVLMs																		
GPT-4o [33]	FPS=3	-	-	32.3	16.7	25.5	22.8	32.8	27.5	37.5	28.6	24.2	19.3	30.8	30.2	20.0	22.0	26.4
Qwen-VL-Max [3]	FPS=3	-	-	33.9	22.4	23.5	21.4	26.2	30.3	41.7	30.2	27.4	26.3	20.0	20.8	16.7	24.0	26.0
Opened-source LVLMs																		
LLaVA-1.5 [25]	4	7B	14.50	32.3	21.3	19.4	17.1	26.2	20.2	36.1	33.3	21.0	21.1	20.0	35.8	16.7	18.0	24.2
InternVL2 [7]	FPS=1	8B	73.23	33.9	24.1	27.6	24.4	**42.6**	33.0	45.8	28.6	19.4	22.8	21.5	34.0	20.0	24.0	28.7
LLaVA-N-Inter [18]	FPS=1	7B	62.78	24.2	23.6	26.5	19.2	31.1	32.1	31.9	17.5	24.2	21.1	26.2	30.2	13.3	20.0	24.4
Qwen2-VL [3]	FPS=1	7B	64.60	27.4	23.0	24.5	23.5	29.5	31.2	47.2	31.7	22.6	28.1	40.0	22.6	30.0	18.0	28.5
LongVA-DPO [54]	128	7B	465.4	35.5	14.9	16.3	19.0	34.4	22.0	37.5	23.8	29.0	22.8	20.0	**37.7**	16.7	12.0	24.4
mPLUG-Owl3 [51]	FPS=1	7B	89.78	30.6	23.6	20.4	22.3	37.7	29.4	48.6	34.9	30.6	24.6	27.7	24.5	13.3	24.0	28.0
LLaVA-N-Video [55]	FPS=1	7B	60.42	31.2	20.2	16.2	17.6	36.5	32.7	30.6	24.5	26.4	24.5	34.7	20.8	20.3	17.0	25.2
Ours	2.36	8.5B	**9.26**	**45.2**	**27.0**	**33.7**	**26.1**	39.3	**36.7**	**55.6**	**41.3**	**30.6**	**29.8**	**30.8**	35.8	**20.0**	**26.0**	**34.1**

Table 1. **Main Result on Our Objective Benchmark.** We report results of closed-source and opened-source LVLMs with ours. The process of constructing task evaluations is shown in the supplementary. TFLOPs refers to the total computational cost of inference, measured under the same 16-second video input.

Models	Log.	Fac.	Acc.	Con.	Overall
Closed-source LVLMs					
GPT-4o	73.15	63.11	61.66	70.02	66.13
Qwen-VL-Max	62.46	50.33	48.43	60.21	53.37
Open-source LVLMs					
LLaVA 1.5	60.53	49.56	49.93	62.1	52.12
InternVL2	70.64	56.32	54.53	66.76	60.05
LLaVA-N-inter	63.27	52.34	48.45	66.78	55.16
Qwen2-VL-7B	66.31	53.67	50.84	68.88	57.66
LongVA-7B-DPO	67.98	54.72	52.78	58.38	57.19
mPLUG-Owl3	66.14	53.05	50.97	67.3	57.14
LLaVA-N-Video	63.42	54.11	49.55	63.31	56.43
Ours	**72.25**	**61.28**	**59.68**	**75.73**	**65.84**

Table 2. **Results on Subjective Benchmark.** We report the metrics of Logic (Log.), Factuality (Fac.), Description Accuracy (Acc.), and Conciseness (Con.).

jective and subjective evaluations, providing a multifaceted analysis of their capabilities.

Evaluation. To more accurately evaluate the open-ended responses of LVLMs, we propose a two-step evaluation method based on fine-grained semantic similarity. In the first step, we assess the semantic similarity between the model's output and the reference answer. If the similarity exceeds 80%, the output is considered potentially correct from a semantic perspective. In the second step, we introduce three highly confounding distractors for each reference answer. We then calculate the similarity between the model's output and each distractor. If the similarity with any distractor surpasses that of the model's output to the reference answer, the response is deemed incorrect. Only if the output passes both steps is it classified as correct.

Meanwhile, we incorporate subjective evaluation by assessing the generated content across multiple dimensions, including logic, factuality, desciption accuracy, and conciseness. To facilitate this, we designed an evaluation framework that prompts GPT-4o [33] to rate the model's output on a scale of 1 to 10 based on the ground truth, along with providing an overall score. Moreover, during evaluation, we do not directly input the options into the model, which effectively prevents potential information leakage. Finally, we report the accuracy for each task across the entire *VideoEspresso* dataset.

5.2. Results on Benchmark

Objective Evaluation Results. We evaluated 7 open-source and 2 closed-source LVLMs on 14 video reasoning tasks in objective evaluation. As shown in Tab. 1, our method achieves the state-of-the-art performance across 12 tasks, and an average accuracy of 34.1%. This performance surpasses that of the top-performing open-source model, InternVL2 [7], and the closed-source GPT-4o [33] by 5.4% and 7.7%, respectively. Compared to our selected backbone, LLaVA-Next-interleave, the performance improves by nearly 10% after fine-tuning with reasoning instructions. In addition to its advantage in video reasoning QA, our method also demonstrates the leading efficiency. Specifically, the average number of input frames is only 1.8% of that used by LongVA-DPO [54], and the FLOPs calculated on the same video input are only 14.74% of those for LLaVA-Next-interleave. Notably, InternVL2 and LongVA-DPO excel in the "Theme Analysis" and "Cooking Process" tasks, which is likely due to their exposure to large size of the same type of data during training process.

Subjective Evaluation Results. We evaluated the quality of LVLMs' answers in subjective evaluation across four aspects, including logic consistency, factuality, accuracy, and conciseness. As shown in Tab. 2, the results align closely with the observations from the objective evaluation. GPT-4o exhibits strong performance in both logical reasoning and factual accuracy, due to its robust language reason-

Setting	Acc.
baseline	34.13
GT-CoT	$72.95_{+38.82}$
w/o Bbox	$33.14_{-0.99}$
w/o CoT	$31.32_{-2.81}$

Table 3. **Ablation Studies on different CoT strategies.** *GT* is 'ground truth'.

Selector	#Frame	Add.	GPU hr.	Inference Memory	Acc.
Uniform	8	-	-	0G + 14G + 40G	33.74
GT	2.98	-	-	0G + 14G +15G	$37.54_{+3.80}$
1B/1.5B	2.77	2.5B	1.33	5G +14G +14G	$34.76_{+1.02}$
1B/0.5B	2.36	1.5B	0.37	3G +14G +12G	$34.13_{+0.29}$

Table 4. **Ablation studies on Selector.** GT refers to the ground truth time of the core frames annotation. "1B/1.5B" represents the selector consists of InternVL2-1B and QwenLM-1.5B, and similarly for "1B/0.5B". Add. stands for additional parameters.

ing capabilities and extensive prior knowledge. However, among all open-source LVLMs, our method outperforms the approaches presented in the table across all four dimensions. Notably, in the Conciseness evaluation, our method surpasses GPT-4o by 5%, further demonstrating the significant contribution of our *VideoEspresso* dataset in enhancing models' ability to learn video reasoning.

5.3. Ablation Study

Ablation on CoT. Our results in Tab. 3 further prove the effectiveness of visual grounding in CoT evidence. Moreover, ablation experiments involving the ground truth of CoT and the ablation of the CoT process further demonstrate the potential of CoT in enhancing performance on visual tasks. The performance boost achieved by the CoT ground truth is significant, highlighting the importance of endowing LVLMs with reasoning QA capabilities.

Ablation on Selector. As shown in Tab. 4, we conducted ablation experiments under different selector combination settings, specifically InternVL2-1B [7] + QwenLM-1.5B [2] and InternVL2-1B + QwenLM-0.5B. In this set of experiments, our core frame selection significantly improved video understanding capabilities compared to the uniform sampling method. Although selectors of different sizes may increase memory usage by 3GB or 5GB, the optimization of redundant keyframe tokens led to a reduction of 26-28GB in memory usage, resulting in a significant improvement in overall video understanding efficiency. We also tested more lightweight LVLMs, such as LLaVA-Next-interleave-0.5B [18] and Qwen-VL-1.5B [3]. However, the results did not meet expectations, likely due to the Tiny LVLMs processing too many image tokens, exceeding their capacity for handling them effectively.

5.4. Adapting Selector to other LVLMs

We further apply the selector to other LVLMs to explore whether the extracted core frames can be effectively generalized to other models in a zero-shot manner. As shown in Tab. 5, we evaluated the method's performance on GPT-4o and several open-source LVLMs. The results demonstrate performance improvements and a reduction in the number of input frames, with the frame input reduced by approximately 15% on both GPT-4o and InternVL2. For the other two models, experiments show that introducing the selector leads to a slight loss in performance, but substantial gains in frame input. Notably, LongVA achieved a 98% reduction in frame input, which highlights that our proposed selector still aids in reducing computational overhead for reasoning in LLMs, as a plug-and-play module.

Model	Sample	#Frame	Ratio_{tok}	TFLOPs	Acc.
GPT-4o	Uniform	16	1	-	26.86
GPT-4o	1B/0.5B	2.77	0.17	-	28.26
GPT-4o	1B/1.5B	2.36	0.15	-	29.45
InternVL2	Uniform	16	1	73.23	28.57
InternVL2	1B/0.5B	2.77	0.17	12.68	29.23
InternVL2	1B/1.5B	2.36	0.15	10.80	30.03
LongVA	Uniform	128	1	465.44	24.41
LongVA	1B/0.5B	2.77	0.02	10.07	23.18
LongVA	1B/1.5B	2.36	0.02	8.58	23.85
LLaVA-N-i	Uniform	16	1	62.78	24.37
LLaVA-N-i	1B/0.5B	2.77	0.17	10.86	24.20
LLaVA-N-i	1B/1.5B	2.36	0.15	9.26	24.26

Table 5. **Evaluations results with selector adoption.**

6. Conclusion

In this paper, we presented *VideoEspresso*, a novel dataset designed to enhance video reasoning by addressing the limitations of existing datasets in terms of scale and granularity. Our approach employs semantic-aware key-frame extraction and leverages GPT-4o to generate fine-grained VideoQA pairs with Chain-of-Thought evidence. By integrating a Hybrid LVLMs Collaboration framework, we achieve cost-effective and accurate video reasoning, outperforming baseline models on the majority of tasks across our proposed benchmark. *VideoEspresso* sets a new starting point in video reasoning, offering rich annotations that facilitate advanced multimodal understanding. We hope our contributions can facilitate future exploration and development of more sophisticated models capable of tackling complex video reasoning challenges.

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